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Technological opportunity discovery based on VERGM and random forest model

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ABSTRACT

Effectively identifying technological opportunities is crucial for R&D organizations to gain a competitive advantage. This paper proposes a novel method to discover technology opportunities in specific fields based on a valued exponential random graph model (VERGM) and random forest models. First, knowledge networks are constructed, and variables reflecting the combination characteristics of knowledge elements are extracted. Then, a valued exponential random graph model is built to identify variables that have significant impacts on the combination frequency of knowledge elements. Furthermore, the identified variables are incorporated into random forest models to train prediction models. Simultaneously, knowledge elements are clustered based on their combination characteristics to identify trending knowledge elements within the field. Finally, technological opportunities for the trending knowledge elements are discovered based on the established prediction models. Patent data in the field of artificial intelligence and the field of Internet of Things is used to verify the effectiveness of the proposed method.

1. Introduction

Technology opportunity discovery refers to the process of identifying and recognizing potential opportunities where technologies can be applied to address existing challenges or create innovative solutions (Yoon et al., 2015). Many existing studies have developed quantitative methods that make use of patent data to analyze the development trend in specific technological fields. In these studies, various variables, such as citations, keywords, technology life cycle, and technology diffusion speed, are used to discover new technology opportunities (Altuntas, Dereli, & Kusiak, 2015; Joung & Kim, 2017; Park & Yoon, 2018; Wang & Hsu, 2023).

A predominant theme in innovation-related literature suggests that innovation emerges by combining and recombining knowledge elements in the processes of investigation and experimentation (Carnabuci & Operti, 2013; Cohen & Caner, 2016; Fleming, 2001; Yayavaram & Ahuja, 2008). A knowledge element refers to a set of tentative conclusions that the research community of a technological field holds about facts, methods, and procedures pertaining to a subject matter (Wang et al., 2014). Two knowledge elements form a tie when they are combined in an invention (Fleming & Sorenson, 2004). Thus, the combination relationships between knowledge elements give rise to knowledge networks (Wang et al., 2014). Some extant studies have proposed methods based on knowledge networks to discover technology opportunities in specific fields (Ren & Zhao, 2021; Shi et al., 2022; Yang et al., 2022). In these studies, various indicators (e.g., technological novelty, technological difficulty, technology importance, and technology potential) are developed to identify new technology opportunities, which are mainly based on knowledge elements' structural characteristics arising from direct ties (e.g., degree centralities, structural holes). While the degree centralities or structural holes of knowledge elements could to some extent reflect their combinatorial opportunities with other knowledge elements (Wang et al., 2014), the combination features of knowledge elements remain under-explored. Specifically, characteristic variables related to indirect ties or tie strengths are seldom considered. In a knowledge network, indirect ties among knowledge elements may also deliver combinatorial opportunities (Guan & Liu, 2016), and a high combination frequency between two knowledge elements is a signal that they are closely related in terms of subject matters and are feasible for deep research (Yayavaram & Ahuja, 2008). Therefore, variables arising from both direct and indirect ties as well as tie strengths should be integrated into a framework to achieve a full understanding of the

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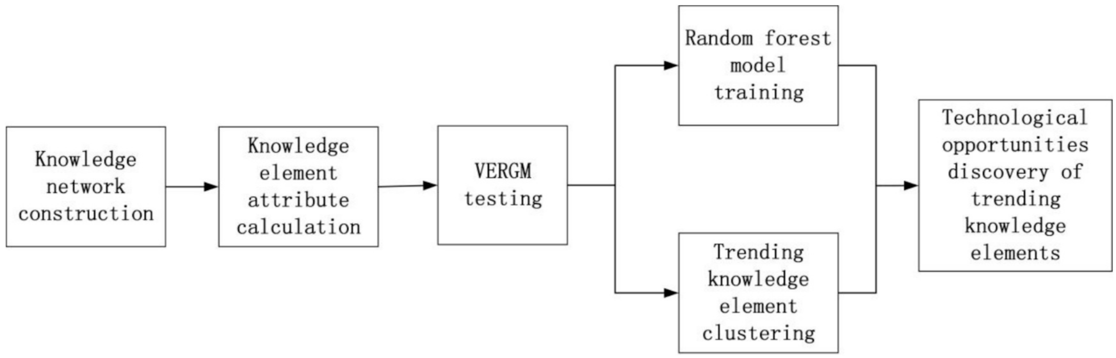

Fig. 1. Technological opportunities discovery processes.

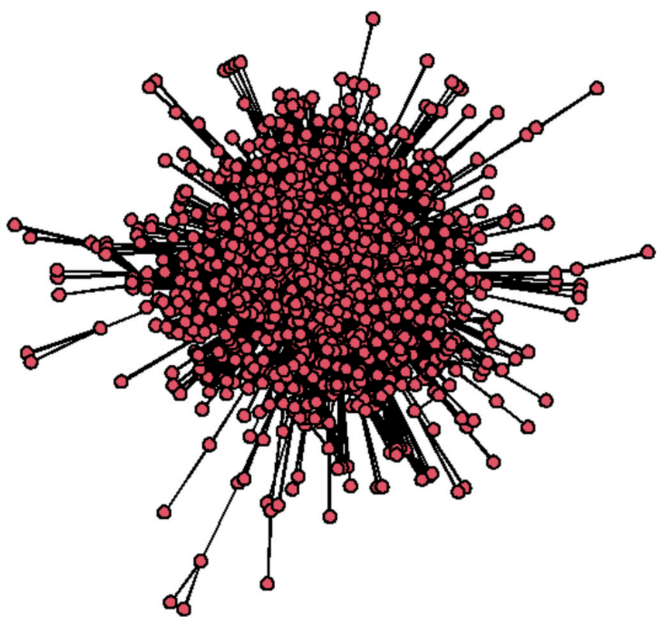

Fig. 2. Knowledge network from 2013 to 2017.

Table 1
VERGM results.

	Estimate	Std.Error
Combination breadth	0.00877***	0.000046
Combination potential	0.00007***	0.000001
Combination intensity	0.01412***	0.000988
Combination distance	-1.25500***	0.000576
nodematch(section)	0.62520***	0.005468
Similarity of neighboring nodes (snn)	29.55000***	0.005885

Note: ***P < 0.001.

Table 2
Model comparison.

Model	R^2	RMSE	MAE
Random forest	0.7641	0.4886	0.2373
LightGBM	0.7433	0.5097	0.2464
Neural network	0.7421	0.5108	0.2726
Support vector machine	0.6899	0.5602	0.2683
Linear regression	0.6606	0.5865	0.3788

combinatorial characteristics of knowledge elements.

In this research, we propose four variables, namely combination breadth, combination potential, combination intensity, and combination distance, to comprehensively reflect knowledge elements' combinatorial features. Combination breadth (i.e., the degree centrality of nodes) has been used in previous studies (Brennecke & Rank, 2017; Wang et al., 2014; Zhang & Luo, 2020a). The combination potential of a knowledge element could be reflected by its betweenness centrality in a knowledge network. Betweenness centrality reflects the importance of a node in connecting other nodes in the network. Knowledge elements with high betweenness centralities are located on the shortest paths between other knowledge elements, playing an important role in connecting the network, and thus obtaining more combination opportunities (Wang et al., 2014). The combination intensity of a knowledge element is reflected by the frequency at which it is combined with other knowledge elements. A high combination frequency between two knowledge elements not only reflects certain technical patterns in the field but also indicates R&D hotspots within the domain (Yayavaram & Ahuja, 2008). The combination distance of a knowledge element is reflected by the shortest distance between that knowledge element and other knowledge elements in the network. A long distance between two knowledge elements in the knowledge network indicates that their connection in terms of subject matters is not close, and the possibility of a combination between them is slim. On the contrary, a short distance between two knowledge elements indicates that there is a great opportunity for a combination between them (Brennecke & Rank, 2017). Based on these characteristic variables, we intend to design a novel approach to identify the combination opportunities of knowledge elements.

More specifically, the combination frequencies between knowledge elements may vary. So does the importance of different knowledge elements in a specific field. Thus, the proposed method aims to identify the potential combination opportunities with high frequencies for trending knowledge elements in the field. To do so, we utilize the valued exponential random graph model (VERGM) to identify factors that have significant impacts on edge weights in a knowledge network (i.e., combination frequencies between knowledge elements), and incorporate them in random forest models to train prediction models. The prediction models are then used to identify important combination opportunities for the trending knowledge elements.

2. Theoretical background

2.1. Technology opportunity discovery

Technology opportunity discovery plays an important role in R&D organizations' technological innovation management. Both qualitative analysis and quantitative methods have been utilized to discover new technology opportunities. In many related studies, patent data is used to predict the development trend of certain technologies. Patent analysis enables the understanding of technology development directions and trends in an effective way (Kim & Lee, 2015; Song, Kim, & Lee, 2018). Some studies use patent data to build knowledge networks, based on which methods for identifying technology opportunities are proposed. For example, Han and Sohn (2016) use association rules to analyze the

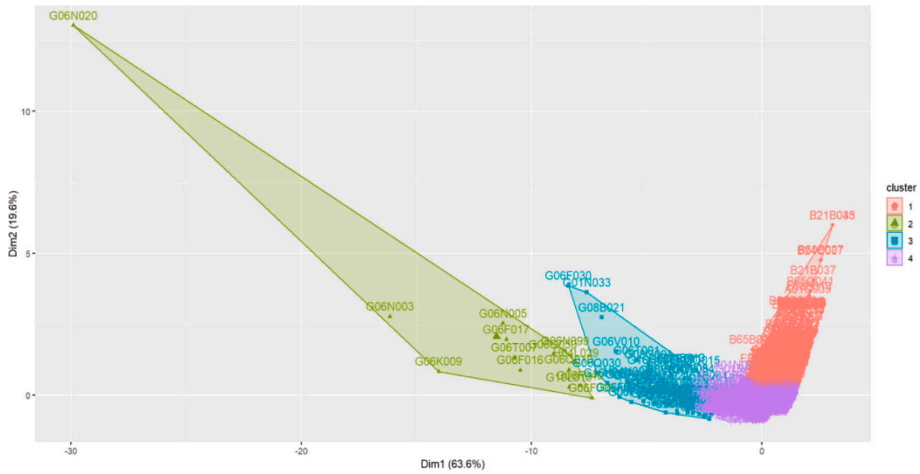

Fig. 3. Clusters of knowledge elements in the field of AI.

Table 3
Means of cluster indicators.

	Combination breadth	Combination potential	Combination distance	Combination intensity
Cluster 1	−0.354	−0.203	1.195	−0.314
Cluster 2	7.959	4.532	−1.782	7.435
Cluster 3	2.193	1.699	−1.235	2.399
Cluster 4	−0.069	−0.069	−0.500	−0.097

combination relationships between knowledge elements in knowledge networks and discover opportunities for technological convergence between the field of information and communication and other technological fields. Zhang and Luo (2020b) construct knowledge networks based on patent data in the field of nanoenergy, and use indicators such as information entropy and combinatorial force to explore technological opportunities in the field. They have identified three important R&D directions. Ren and Zhao (2021) construct domain knowledge networks based on patent data and use regression analysis to obtain variables that significantly affect the value of technology opportunities. The variables are then used to form an objective function for searching and interpreting optimal opportunities. Kim et al. (2021) analyze the development trajectory of blockchain technology based on patent knowledge networks and identify technological opportunities in five subfields based on the knowledge flow of main paths.

2.2. Valued exponential random graph model

In a large number of studies on the formation mechanism of social networks, logistic regression models are commonly used. However, conventional regression models require assumptions of independence between observations and fail to incorporate endogenous structural effects of the observed network (Kim et al., 2016). To solve this problem, scholars propose exponential random graph models (ERGM). ERGM can integrate both endogenous and exogenous variables to analyze the formation of networks. Formula (1) presents the general form of ERGM (Van der Pol, 2019).

$$P(\mathrm{Y} = y) = (1/k)\exp\{\sum \theta_A Z_A(y)\} \quad (1)$$

Table 4
Technological opportunity prediction for trending knowledge elements.

G06Q050		**G10L015**		**G06N099**		**G06F040**		**G06F003**		**G06Q010**		**G06N003**	
CO	**CV**	**CO**	**CV**	**CO**	**CV**	**CO**	**CV**	**CO**	**CV**	**CO**	**CV**	**CO**	**CV**
H04R003	0.945	G01S007	0.957	G06V010	0.920	G05B015	0.970	H04W064	0.941	G10L021	0.971	H04W076	0.939
G01S007	0.924	G06T015	0.939	H04W084	0.908	G06T015	0.969	G01B011	0.920	H04R003	0.934	G06K017	0.918
G16B040	0.920	G06V030	0.924	B60W060	0.860	G02B027	0.947	H04N019	0.917	G02B027	0.917	H04H060	0.912
H03M007	0.920	G01N021	0.918	G06K019	0.733	G08C017	0.944	H04W024	0.913	G06V040	0.916	A61B017	0.903
H04B007	0.898	G01B011	0.910	G16B030	0.729	G01N033	0.938	H04B001	0.911	G09G005	0.915	A61F009	0.894
H04W072	0.894	G16H030	0.906	A61H003	0.726	G08B013	0.931	H04B007	0.907	B60R016	0.913	E21B044	0.891
A61B090	0.888	H04W064	0.905	G06F018	0.724	G07C009	0.930	H04B010	0.901	H01L021	0.887	G07G001	0.885
H04L043	0.887	G06F012	0.901	H04L005	0.723	G11B027	0.930	B60R025	0.896	A61B001	0.883	F21V023	0.884
H04N019	0.886	G06T017	0.897	F21V033	0.722	G06T009	0.927	H04L001	0.893	H04R001	0.882	H01M010	0.883
H04N009	0.883	A61B006	0.892	H02J013	0.713	G06T017	0.925	E21B047	0.893	G06F012	0.875	G08B019	0.871
G06F017		**G06F016**		**G06K009**		**G06T007**		**H04L029**		**G06N020**		**G06N005**	
CO	**CV**	**CO**	**CV**	**CO**	**CV**	**CO**	**CV**	**CO**	**CV**	**CO**	**CV**	**CO**	**CV**
G08C017	0.984	G01S007	0.926	G08B027	0.941	G08C017	0.984	G06T015	0.968	B64D047	0.900	G08C017	1.011
B60R011	0.942	B60R011	0.925	B22F003	0.935	G05B023	0.924	G02B027	0.939	B60K035	0.889	B60R011	0.935
G05B015	0.942	G01S017	0.923	H05K007	0.932	H04W012	0.904	G06V040	0.933	G10K011	0.888	G01D021	0.924
G08B013	0.925	G08B013	0.919	H04L005	0.916	H04W084	0.899	B25J019	0.920	B65G001	0.842	G01S017	0.915
G01S007	0.921	H04W088	0.916	G16B005	0.914	G06F012	0.883	G01S007	0.919	H05K007	0.837	G08B007	0.911
A61B008	0.914	G05B015	0.913	C02F003	0.884	F24F011	0.852	G06T013	0.911	A61H003	0.819	H04N013	0.902
H04W072	0.911	G01S005	0.906	H05K013	0.879	G10L019	0.843	G01N033	0.909	G02B006	0.815	G07F017	0.895
G06K017	0.902	G07F017	0.902	G01M017	0.875	H02J007	0.793	B60W060	0.906	F21V033	0.793	G06T009	0.890
G01T001	0.901	B60W040	0.902	B23Q015	0.872	H04W024	0.782	G01N021	0.898	G07F011	0.721	G01S013	0.888
B25J019	0.898	H04W008	0.900	H04W056	0.865	H04Q009	0.774	B60R011	0.893	G02B026	0.700	G11C011	0.878

Note: CO: Combination opportunity; CV: Combination Value.

Table 5
Technological opportunity validation for trending knowledge elements.

G06Q050			G10L015			G06N099			G06F040			G06F003			G06Q010			G06N003		
CO	**PF**	**AF**	**CO**	**PF**	**AF**	**CO**	**PF**	**AF**	**CO**	**PF**	**AF**	**CO**	**PF**	**AF**	**CO**	**PF**	**AF**	**CO**	**PF**	**AF**
H04R003	8	4	G01S007	8	2	G06V010	7	7	G05B015	8	9	H04W064	8	9	G10L021	8	6	H04W076	8	8
G01S007	7	8	G06T015	8	9	H04W084	7	7	G06T015	8	8	G01B011	7	7	H04R003	8	4	G06K017	7	9
G16B040	7	9	G06V030	7	7	B60W060	6	0	G02B027	8	5	H04N019	7	7	G02B027	7	8	H04H060	7	13
H03M007	7	7	G01N021	7	2	G06K019	4	5	G08C017	8	0	H04W024	7	8	G06V040	7	11	A61B017	7	5
H04B007	7	5	G01B011	7	0	G16B030	4	0	G01N033	8	3	H04B001	7	10	G09G005	7	10	A61F009	7	9
H04W072	7	0	G16H030	7	7	A61H003	4	2	G08B013	8	7	H04B007	7	6	B60R016	7	9	E21B044	7	8
A61B090	7	6	H04W064	7	3	G06F018	4	5	G07C009	8	4	H04B010	7	7	H01L021	7	10	G07G001	7	8
H04L043	7	8	G06F012	7	5	H04L005	4	3	G11B027	8	9	B60R025	7	7	A61B001	7	5	F21V023	7	1
H04N019	7	4	G06T017	7	7	F21V033	4	0	G06T009	7	8	H04L001	7	12	H04R001	7	7	H01M010	7	7
H04N009	7	7	A61B006	7	1	H02J013	4	3	G06T017	7	9	E21B047	7	6	G06F012	6	8	G08B019	6	8
G06F017			**G06F016**			**G06K009**			**G06T007**			**H04L029**			**G06N020**			**G06N005**		
CO	**PF**	**AF**	**CO**	**PF**	**AF**	**CO**	**PF**	**AF**	**CO**	**PF**	**AF**	**CO**	**PF**	**AF**	**CO**	**PF**	**AF**	**CO**	**PF**	**AF**
G08C017	9	10	G01S007	7	6	G08B027	8	9	G08C017	9	9	G06T015	8	8	B64D047	7	9	G08C017	9	7
B60R011	8	0	B60R011	7	9	B22F003	8	6	G05B023	7	10	G02B027	8	6	B60K035	7	9	B60R011	8	3
G05B015	8	7	G01S017	7	8	H05K007	8	8	H04W012	7	7	G06V040	8	5	G10K011	7	7	G01D021	7	7
G08B013	7	4	G08B013	7	8	H04L005	7	8	H04W084	7	7	B25J019	7	5	B65G001	6	7	G01S017	7	10
G01S007	7	10	H04W088	7	5	G16B005	7	11	G06F012	7	7	G01S007	7	6	H05K007	6	9	G08B007	7	6
A61B008	7	8	G05B015	7	10	C02F003	7	7	F24F011	6	0	G06T013	7	8	A61H003	6	9	H04N013	7	9
H04W072	7	7	G01S005	7	4	H05K013	7	0	G10L019	6	10	G01N033	7	8	G02B006	6	3	G07F017	7	7
G06K017	7	6	G07F017	7	8	G01M017	6	5	H02J007	5	7	B60W060	7	7	F21V033	5	5	G06T009	7	7
G01T001	7	7	B60W040	7	9	B23Q015	6	5	H04W024	5	6	G01N021	7	4	G07F011	4	0	G01S013	7	6
B25J019	7	8	H04W008	7	3	H04W056	6	0	H04Q009	5	3	B60R011	7	2	G02B026	4	3	G11C011	7	4

Note: CO: Combination opportunity; PF: Predicted Frequency; AF: Actual Frequency.

Where k is a constant to ensure that the probability of network structure y is between 0 and 1, θ_A is the coefficient of the statistic $Z_A(y)$. $Z_A(y)$ could be endogenous network structures or exogenous variables.

Traditional ERGMs do not consider the weights of edges in a network, and are thus unable to analyze the mechanisms that influence the interaction intensity between nodes in the network. To address this issue, researchers put forth valued exponential random graph models (VERGM), which incorporate the weights of edges into the analysis. Formula (2) presents the general form of VERGM.

$$P(Y = y) = (1/k)h(y)\exp\{\sum \theta_A Z_A(y)\} \quad (2)$$

Where h(y) is the reference distribution of the weights of edges. VERGM has been used in the study of the impact mechanisms of various networks, such as policy networks (Scott, 2016), international R&D cooperation networks (Whetsell, 2023), R&D personnel communication networks (Somik, 2020), and global civil society networks (Saffer, Pilny, & Sommerfeldt, 2022).

2.3. Random forest

A decision tree is one of the frequently used classifiers. A random forest (RF) consists of a collection of decision trees, which results in a reduction of variance compared to a single decision tree (Couronne, Probst, & Boulesteix, 2018). In previous classification studies, RF is reported as one of the foremost classifiers for producing high accuracy (Phan Thanh & Kappas, 2018). Each tree in a random forest is grown with a subset of data made from bootstrap and a random subset of variables (Breiman, 2001). To classify a new case, RF puts the new case down each tree in the forest to get a vote for classification and chooses the classification with the most votes. Because of the bootstrap, the samples involved in the model construction account for two-thirds of the training set. The remaining one-third is called Out-Of-Bag (OOB) data, which can be used to assess the performance of the constructed model by an unbiased internal cross-validation (Qiu, et al., 2020).

3. Methodology

This research aims to discover potential pairs of knowledge elements with high combination frequencies. We need to predict the combination frequencies of knowledge elements that have not yet been combined. Specifically, we take the steps shown in Fig. 1 to identify technological opportunities in a specific field.

Step 1: We extract knowledge elements from patent data and construct a knowledge network based on their combination relationships in patents. Following previous studies (Brennecke & Rank, 2017; Guan & Liu, 2016; Zhang & Luo, 2020a), we adopt the International Patent Classification (IPC) codes to approximate knowledge elements. The IPC code is a hierarchical classification system that consists of sections, classes, subclasses, main groups, and subgroups. Many previous studies use subclass level IPC codes (the first four digits in IPC codes) to represent knowledge elements. However, subclass level IPC codes may embrace a too wide range of heterogeneous technologies. By comparison, main group level IPC codes are more fine-grained and accordingly more appropriate to reflect specific technological processes and mechanisms (Park & Yoon, 2017). Thus, we use main group level IPC codes to denote knowledge elements. Nodes in a knowledge network represent knowledge elements. If two knowledge elements co-occur in a patent, a tie is established between them in the knowledge network.

Step 2: We calculate the combination characteristics of knowledge elements based on the established knowledge network. Specifically, we calculate four combinatorial characteristic variables, namely combination breadth, combination potential, combination intensity, and combination distance.

The combination breadth of a knowledge element is reflected by its degree centrality in the knowledge network (Wang et al., 2014).

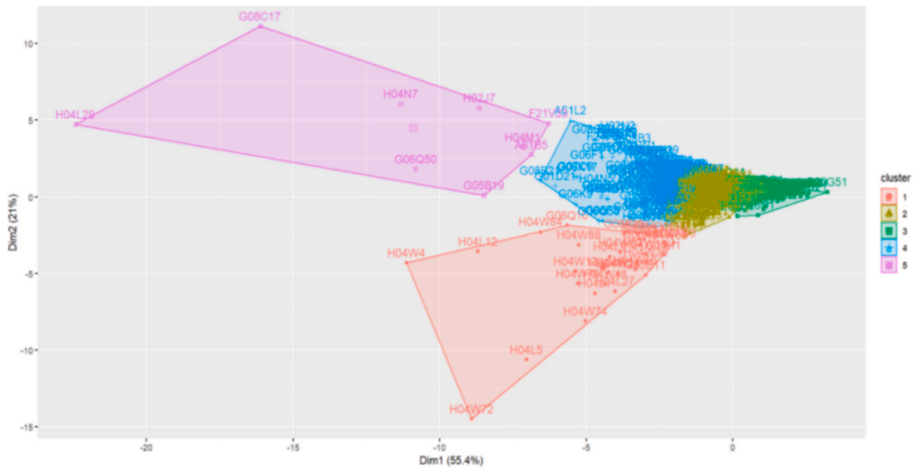

Fig. 4. Clusters of knowledge elements in the field of IoT.

$$\text{breadth}_i = \text{degree}_i \tag{3}$$

Where breadth_i denotes the combination breadth of knowledge element i, degree_i represents its degree centrality (i.e., the number of direct ties it has) in the knowledge network.

The combination potential of a knowledge element is calculated by the following formula:

$$\text{cp}_i = \sum\nolimits_{s\neq i\neq t} \frac{n^i_{st}}{g_{st}} \tag{4}$$

where cp_i denotes the combination potential (i.e., betweenness centrality) of knowledge element i, g_{st} represents the number of shortest paths from node s to node t, ni st represents the number of shortest paths from node s to node t that pass through node i.

The combination intensity of a knowledge element is calculated as follows:

$$\text{intensity}_i = \frac{\sum_{j=1}^{n} \text{Cf}_{ij}}{n} \tag{5}$$

where intensity_i denotes the combination intensity of knowledge element i, n represents the number of adjacent nodes of knowledge element i, and Cf_{ij} represents the combination frequency of knowledge element i and the jth connected knowledge element.

The combination distance of a knowledge element is calculated by the following formula:

$$\text{distance}_i = \frac{\sum_{i\neq j}^{N-1} d_{ij}}{N-1} \tag{6}$$

where distance_i denotes the combination distance of knowledge element i, N is the number of nodes in the knowledge network, and d_{ij} represents the shortest distance between knowledge element i and knowledge element j in the knowledge network.

Step 3: We construct a VERGM and perform parameter estimation to identify factors that have significant impacts on edge weights (i.e., combination frequencies between knowledge elements) in the knowledge network. In this study, we integrate knowledge elements' combination characteristics and their dyadic characteristics to construct the following VERGM:

$$\begin{aligned} P(Y=y) = &(1/k)h(y)\exp\{\theta_1 nodecov(''breadth'') + \theta_2 nodecov(''cp'') \\ &+ \theta_3 nodecov(''intensity'') + \theta_4 nodecov(''distance'') \\ &+ \theta_5 nodematch(''section'') + \theta_6 edgecov(snn)\} \end{aligned} \tag{7}$$

where $\theta_1 \sim \theta_6$ is the parameter estimation for each statistic. In the equation, the "nodecov" function measures the effect of nodes' attributes on the weights of edges. Specifically, nodecov("breadth"), nodecov("cp"), nodecov("intensity"), and nodecov("distance") are used to test the effects of the combination breadth, combination potential, combination intensity, and combination distance of knowledge elements on edge weights. The "nodematch" function measures homophily effects on edge weights, assessing whether nodes with matching attributes form stronger connections. For instance, nodematch("section") is used to test whether knowledge elements of the same IPC section have higher combination frequencies. The "edgecov" function includes edge-level covariates to explain variation in edge weights. Specifically, edgecov (snn) is to examine the effects of transitivity on edge weights, which refers to the impact of common neighboring nodes of two knowledge elements on their edge weights. snn represents the similarity of neighboring nodes between two knowledge elements. It is calculated as follows:

$$\text{snn}_{ij} = \frac{f_i \cap f_j}{f_i \cup f_j} \tag{8}$$

where snn_{ij} represents the similarity of neighboring nodes between knowledge element i and knowledge element j. f_i and f_j represent the sets of neighboring nodes of knowledge elements i and j, respectively. $f_i \cap f_j$ represents the number of common neighboring nodes between the two knowledge elements, and $f_i \cup f_j$ represents the total number of their neighboring nodes.

Step 4: The variables that are found to have significant impacts on edge weights in the third step are taken as indicators and input into a random forest model for training and building a prediction model. Simultaneously, based on these variables, clustering of knowledge elements in the network is performed to discover trending knowledge elements within the field. Trending knowledge elements represent new developments, hot topics, or important insights in a particular field. Generally, we regard knowledge elements with high combination breadth, combination potential, combination intensity, and low combination distance as trending knowledge elements, because they exhibit characteristics that align with centrality, adaptability, integration, and influence within the knowledge network. Specifically, high combination breadth signals cross-disciplinary relevance (Wang et al., 2014). Trending topics often emerge at the intersection of fields. High combination potential indicates these elements are primed to form new connections with other nodes. They act as "bridges" for ideas, enabling novel combinations that drive emerging trends (Brennecke & Rank, 2017). High combination intensity reflects active engagement—frequent use of the knowledge element (Luo, Qu, & Cheng, 2024). Trending topics are often "hot" because they are frequently discussed, tested, or applied. Low combination distance indicates these knowledge elements are centrally positioned in the network, acting as hubs for knowledge exchange (Guan & Liu, 2016). Trending topics are rarely isolated; they are tightly linked to foundational or emerging ideas. Together, these

Table 6
Robust check results.

H04M1				F21V33				H04L29				B23K35				G06Q50			
CO	**CV**	**PF**	**AF**	**CO**	**CV**	**PF**	**AF**	**CO**	**CV**	**PF**	**AF**	**CO**	**CV**	**PF**	**AF**	**CO**	**CV**	**PF**	**AF**
H04J11	0.959	8	7	F24F7	0.993	9	9	B23Q1	0.867	6	1	B23K35	0.824	6	0	G03H1	0.920	7	8
A01N65	0.869	6	0	F16M11	0.895	7	10	A41B9	0.848	6	3	B23Q15	0.811	5	3	A47B37	0.877	7	7
B26D7	0.856	6	6	F24D19	0.831	6	6	A44B18	0.848	6	0	B23Q11	0.794	5	6	H02N2	0.848	6	5
B60L8	0.838	6	2	F03B13	0.810	5	5	A62B7	0.845	6	4	G01G21	0.791	5	7	B02C23	0.843	6	6
B66F11	0.834	6	0	F16M13	0.810	5	5	B25B23	0.842	6	3	G03H1	0.765	5	3	B08B3	0.836	6	6
H04W36	0.814	6	4	F22B37	0.807	5	2	G01L25	0.842	6	0	G01K3	0.764	5	5	A44B1	0.825	6	0
H02N2	0.806	5	5	G01H1	0.797	5	0	B61D33	0.837	6	1	E04B7	0.750	5	2	H04R9	0.823	6	1
B01D46	0.805	5	5	F21V5	0.797	5	7	G01B17	0.824	6	6	B08B3	0.750	5	4	B05C5	0.821	6	2
H04L61	0.792	5	4	A01G31	0.786	5	4	A61F13	0.779	5	5	A62B7	0.748	5	5	B05C11	0.821	6	7
H04L41	0.792	5	6	F16K37	0.779	5	6	E01C23	0.768	5	4	F21V15	0.743	5	6	A01G23	0.815	6	6
G05B19				**H04N7**				**H02J7**				**A61B5**							
CO	**CV**	**PF**	**AF**	**CO**	**CV**	**PF**	**AF**	**CO**	**CV**	**PF**	**AF**	**CO**	**CV**	**PF**	**AF**				
B26D7	0.879	7	7	H02N2	0.854	6	2	H04W60	0.927	7	2	B01D46	0.877	7	7				
G01G21	0.875	7	6	H04L41	0.852	6	7	H04L25	0.870	6	7	B08B3	0.868	6	4				
G01N29	0.807	5	7	B23K35	0.844	6	0	H04W36	0.857	6	6	C02F1	0.789	5	5				
F26B21	0.807	5	4	B23P23	0.825	6	2	H04L27	0.834	6	8	H04W60	0.787	5	1				
G03H1	0.797	5	3	H04L43	0.821	6	6	B26D7	0.825	6	8	A47D13	0.786	5	5				
H01B9	0.786	5	0	F01D15	0.812	5	0	F16K5	0.798	5	3	H02S20	0.783	5	4				
B23Q1	0.783	5	5	H04L61	0.797	5	7	B23P23	0.792	5	1	H04W36	0.782	5	6				
B61D33	0.775	5	0	A62B7	0.778	5	5	B23Q15	0.791	5	0	H04W80	0.753	5	5				
A63F11	0.770	5	3	H02K41	0.778	5	3	B23Q11	0.790	5	6	A63F3	0.751	5	4				
A62B7	0.766	5	5	F21V15	0.772	5	6	A61F5	0.788	5	6	E05F15	0.749	5	3				

Note: CO: Combination opportunity; CV: Combination Value; PF: Predicted Frequency; AF: Actual Frequency.

metrics describe knowledge elements that are widely connected (breadth), actively used (intensity), poised for growth (potential), and centrally embedded (low distance)—key indicators of trendiness in a specific field.

Step 5: Based on the random forest prediction model yielded in Step 4, the potential combinations of trending knowledge elements are calculated, and the combinations with high values are identified as new technological opportunities.

4. Empirical analysis

4.1. Data

This study conducts an empirical analysis using patent data in the field of artificial intelligence (AI). From the perspective of domain knowledge characteristics, the field of AI covers multiple disciplines such as computer science, physics, biology, mathematics, and cognitive science. The feature of interdisciplinary knowledge integration in the field makes it a suitable case for analyzing knowledge combinations. From a strategic perspective, many countries consider the development of AI as an important strategy and a crucial path to gain future competitive advantages. Therefore, it is practically significant to identify technological opportunities in the field of AI. Following the search strategy used by Zhang and Luo (2020a), we retrieve patent data in the field of AI between 2013 and 2021 from the Derwent patent database. After data cleaning, a total of 60,927 records are obtained.

4.2. Random forest model construction and training

4.2.1. Model indicators

As is shown in Fig. 1, we employ VERGM to identify factors that significantly affect edge weights (i.e., combination frequencies of knowledge elements). We first build a knowledge network according to the combination relationships of knowledge elements in patents from 2013 to 2017. Fig. 2 presents the knowledge network that consists of 1195 nodes and 9773 edges. Degree centralities of the nodes range between 1 and 432, with a mean of 16.4. Edge weights range from 1 to 446, with a mean of 95.6. These data indicate that nodes in the knowledge network vary in terms of combination features. Based on these differences, we can identify the trending knowledge elements and explore technological opportunities involving them. Based on the constructed knowledge network and the formulas mentioned above, we calculate the variables included in equation (7), i.e., combination breadth, combination potential, combination intensity, combination distance, the section homophily of knowledge elements, and the similarity of neighboring nodes. Regarding the sections of knowledge elements, the IPC system divides IPC into 8 sections. The first letter of an IPC code reflects the section to which the knowledge element belongs. In this study, we use this classification to divide the knowledge elements into 8 categories for testing the homophily effect (1 if both knowledge elements belong to the same section, 0 otherwise).

We use the ERGM and ERGM.COUNT packages in R to perform parameter estimation for the VERGM shown by equation (7), where the Markov Chain Monte Carlo (MCMC) estimation method is adopted. Table 1 presents the results. As is shown in Table 1, the coefficients of the four characteristic variables, homophily effect, and the similarity of neighboring nodes are all significant ($P < 0.001$), indicating that these variables or effects have significant impacts on the edge weights in the knowledge network. Thus, we incorporate these indicators into random forest models for training.

4.2.2. Model training

To determine the number of training samples, we first count the newly-emerged combinations of 1195 knowledge elements from 2018 to 2019 (i.e., knowledge element combinations that did not occur between 2013 and 2017), and identify a total of 8265 new combinations. Since the number of knowledge element pairs that are not combined during 2018–2019 is much greater than the number of pairs that are combined, to ensure a similar number of knowledge element pairs in the training set for both scenarios, we randomly select 8265 pairs of knowledge elements that are not combined, and finally obtain a dataset of 16,530 samples.

Since each sample consists of a pair of knowledge elements, the interaction terms of the combination characteristic variables of the two knowledge elements (e.g., the interaction of the combination breadth of the two knowledge elements), the similarity of neighboring nodes, the shortest distance between the two knowledge elements, and the homophily are included in the training model as independent variables. In terms of the dependent variable, the random forest model can handle both categorical and continuous variables. For categorical variables, a classification model is used, while for continuous variables, a regression model is operated. We first train the classification model to predict whether two knowledge elements would be connected, and then train the regression model to predict the combination frequencies of two knowledge elements. For the classification model training, 2/3 of the data is randomly selected from the dataset as the training set, and the remaining 1/3 is used as the test set (Breiman, 2001; Qiu, Bao, Yang, Wang, & Jia, 2020). For the random forest regression model training, the combination frequency of two knowledge elements that occur between 2018 and 2019 is used as the dependent variable. Since the values of the samples are right-skewed, we apply a log transformation after adding 1 to the values.

When building a random forest model, two critical parameters named mtry and ntree should be set, which could affect model accuracy (Maruf, Javed, & Babri, 2016). mtry is the number of variables used to split a decision tree, and ntree is the number of decision trees to be generated in the model. Following previous research (Qiu et al., 2020), mtry is first determined according to the lowest OOB error. Then, ntree is chosen based on the lowest mean squared error and the selected mtry. Finally, the value of mtry is set to 5, and the value of ntree is set to 200. After determining the parameters, both models are trained. In terms of the classification model, the model obtained from the training set is applied to the test set. It achieves an accuracy of 93.21 %, a precision of 93.78 %, a recall of 92.56 %, and an F1-score of 93.17 %. These indicators suggest that the classification model has a good predictive performance. In terms of the regression model, the variance explained by the model (R-square) is 76.41 %, indicating a good fit. To further evaluate the performance of the random forest model, we also run the data using some other commonly-used models, such as lightGBM, neural network, support vector machine, and linear regression models. Table 2 presents the results. It can be seen that the random forest model has the highest R-square value, as well as the lowest RMSE (Root Mean Squared Error) and MAE (Mean Absolute Error) values, indicating that the random forest model has a superior performance in predicting the combination frequencies between knowledge elements.

4.3. Trending knowledge elements identification

To identify trending knowledge elements in the field, we use combination breadth, combination potential, combination intensity, and combination distance as indicators and conduct a clustering analysis. Based on the patent data from 2018 to 2019, we construct a knowledge network that consists of 1731 nodes, and calculate the combination characteristic variables for each knowledge element in the knowledge network. Then, we used the K-means clustering method to cluster the knowledge elements. The K-means clustering method is scalable and efficient for clustering large sample data. Before running the K-means clustering method, the number of clusters needs to be determined. By calculating the silhouette coefficients for different numbers of clusters, we find that the silhouette coefficient is the highest when the number of clusters is 4. Therefore, the knowledge elements are divided into 4 clusters (Kim & Bae, 2017). The clustering results are shown in Fig. 3. This figure is generated by the "fviz_cluster" function in R, which reduces dimensionality for visualization using principal component analysis. It automatically projects high-dimensional data onto the first two principal components (PC1 and PC2), which capture the maximum variance in the data. The figure displays clusters in this reduced space, with axes labeled as "Dim1" (PC1) and "Dim2" (PC2).

Table 3 presents the means of clustering indicators for the 4 clusters. It can be seen from the table that the second cluster has the highest combination breadth, combination potential, and combination strength, as well as the lowest combination distance. High combination breadth, high combination potential, and low combination distance indicate promising combination opportunities. High combination strength reflects technical patterns and R&D hotspots. Thus, knowledge elements in this cluster could represent the R&D trends in the field. We refer to these knowledge elements as the trending knowledge elements. In terms of specific subject matters, the 14 knowledge elements in the second cluster reflect some R&D hotspots and trends in the field of AI: text or image recognition (G06K009, G06T007), computer systems based on specific computational models such as machine learning models and biological models (G06N003, G06N005, G06N020, G06N099), electric digital data processing such as data communication, information retrieval, and natural language processing (G06F003, G06F016, G06F017, G06F040, G06Q010, G06Q050, H04L029), and speech recognition (G10L15).

Additionally, there are 553 nodes in Cluster 1. As shown in Table 3, nodes in this cluster have low combination breadth, combination potential, and combination intensity, but high combination distance. These features indicate that the 553 nodes are not closely related to other nodes in the knowledge network in terms of subject matters. Cluster 3 contains 73 nodes, which present medium combination breadth, potential, distance, and intensity. These attributes suggest that the 73 nodes play intermediary roles that connect different parts of the knowledge network but lack strong integration into central hubs. Cluster 4 contains the largest number of nodes (1091 nodes), which have relatively low combination breadth, potential, and intensity, but relatively high combination distance. These nodes may represent specialized or niche subject matters with limited interaction outside their immediate domains.

4.4. Technological opportunities discovery for trending knowledge elements

We employ the previously trained random forest regression model to discover the combination opportunities for the 14 trending knowledge elements. We compute the combination characteristic variables of each knowledge element based on the knowledge network for 2018–2019. Then, we screen the knowledge elements that are not combined with the trending knowledge elements during this period. The yielded knowledge elements are paired with the trending knowledge elements. The related indicators of each knowledge element pair are first input into the established random forest classification model to check whether they would be combined. The knowledge element pairs with a classification result of being combined are then input into the random forest regression model to calculate combination values. The top 10 knowledge element pairs with the highest combination values are viewed as important technological opportunities in the field. Specifically, the combination value (CV) for a knowledge element pair in random forest regression is computed as the average of the predictions from all individual decision trees in the ensemble:

$$\mathrm{CV} = \frac{1}{\mathrm{B}} \sum_{\mathrm{b}=1}^{\mathrm{B}} \mathrm{T_b}(\mathrm{x}) \tag{9}$$

Where x is the input feature vector that consists of the interaction terms of the combination characteristic variables as well as the dyadic variables mentioned above. B is the total number of decision trees in the forest. $T_b(x)$ is the prediction of the b-th decision tree for input x, which is computed through a tree traversal based on learned split rules, ultimately outputting a value stored in the leaf node.

The potential opportunities of the 14 trending knowledge elements are shown in Table 4. The first column below each trending knowledge element lists the knowledge elements that have potential combination relationships with it, and the second column lists the corresponding combination values.

The above combinations are likely to occur frequently in subsequent periods. Thus, they can be integrated to identify technological opportunities. Specifically, three types of opportunities can be explored: technology enhancement opportunities, scenario expansion opportunities, and technology replacement opportunities. Technology enhancement refers to the improvement of existing technology combinations through performance optimization or functional upgrades. Scenario expansion involves the adaptation of the technology combinations to new domains or environments. Technology replacement refers to the displacement of legacy technical solutions by emerging technology combinations that address fundamental flaws (e.g., cost inefficiency, scalability limits, security vulnerabilities).

Taking the trending knowledge element G10L015 as an example, it represents technologies related to speech analysis or synthesis, and speech recognition. From the perspective of technology enhancement, technological opportunities may exist in autonomous driving, where voice queries trigger radar (G01S007) and camera (G06V030) data fusion, enabling AI to generate a robust environmental model. In terms of scenario expansion, there may be potential technological opportunities in developing voice-controlled imaging workflows that are integrated with smart diagnostic tools (A61B006), which address the problem that surgeons need hands-free control of medical imaging systems. From the perspective of technology replacement, researchers may explore technological opportunities in industrial inspections, where workers use voice commands to control spectrometers in noisy factory scenarios, with 5G edge networks (H04W064) enabling real-time AI analysis and voice feedback.

For another example, regarding technological opportunities involving knowledge element G06N003 (computer systems based on neural networks), researchers may explore technological opportunities in deploying neural networks to analyze real-time drilling data (E21B044) and optimize energy use via battery management (H01M010), reducing equipment wear and improving efficiency (technology enhancement). From the perspective of scenario expansion, researchers may integrate neural networks with video analysis (A61B017) to assist surgery operations. In terms of technology replacement, researchers may develop edge-deployed neural networks to detect abnormality and trigger alerts via IoT security systems (G08B019).

4.5. Validation of results

To verify the accuracy of the predicted results, we count the actual combination frequencies of the knowledge element pairs between 2020 and 2021. The results are shown in Table 5. The numbers in the second column below each trending knowledge element are the predicted combination frequencies, which are converted from the combination values listed in Table 3 (the inverse operation of the previous logarithmic transformation), and the numbers in the third column are the actual combination frequencies. It can be seen that, from the perspective of whether combinations occur, the average prediction accuracy for the 14 trending knowledge elements is 92 %, with a maximum of 100 % and a minimum of 70 %. Thus, the method we propose can effectively discover technological opportunities for the trending knowledge elements. In terms of combination frequencies, the highest prediction accuracy is 83.04 %, and the lowest is 55 %. The average prediction accuracy reaches 69.1 %, indicating that the proposed method has good prediction performance.

Table 5 also shows that the knowledge elements combined with the trending knowledge elements belong to multiple categories. Therefore, the proposed method can provide useful R&D directions and guidance for researchers in various fields. The examples (i.e., potential opportunities) discussed in section 4.4 have turned into real applications, such as patent WO2021258240 “Voice or speech recognition in noisy environments” (G10L015 + H04W064), EP21177923 “Camera tracking bar for computer assisted navigation during surgery” (G10L015 + A61B006), KR102777694B1 “Method and apparatus for silent speech recognition using radar” (G10L015 + G01S007), CN202110360348.1 “Life cycle cost and battery temperature optimization method based on electric vehicle battery” (G06N003 + H01M010), US17117730 “Automated instrument or component assistance using mixed reality in orthopedic surgical procedures” (G06N003 + A61B017), and CN202010514752.5 “A kind of pipe gallery abnormal detection system and method” (G06N003 + G08B019). These patents can further validate the effectiveness of the proposed method.

4.6. Robustness check

To check the robustness of the proposed method, we collect patent data in the field of Internet of Things (IoT) from 2010 to 2021 from the Incopat patent database. A total number of 88,767 patents are obtained. Meanwhile, to examine whether the division of time periods has an impact on the prediction of technological opportunities, we adopt a new time division. Specifically, based on patent data from 2010 to 2015, we construct a knowledge network and calculate the combination characteristic variables of knowledge elements. Random forest models are trained based on the variables yielded and the combinations of knowledge elements from 2016 to 2018. The trending knowledge elements are identified based on the knowledge network from 2016 to 2018. Finally, we validate the prediction accuracy of technology opportunities for the trending knowledge elements using patent data from 2019 to 2021. Fig. 4 presents the result of cluster analysis, where nine trending knowledge elements are identified in cluster 5.

Table 6 shows the prediction results. It can be seen that, in terms of combination occurrence, the average accuracy of the predicted combination opportunities for the trending knowledge elements is 86.7 %, and the average prediction accuracy for the combination frequencies between the knowledge element pairs is 64.2 %. Thus, the proposed method could be considered robust.

5. Discussion and conclusion

Based on knowledge combination theory, this research uses VERGM to identify factors that significantly affect the combination frequencies between knowledge elements, and constructs random forest models to discover the technological opportunities of trending knowledge elements in specific fields. The method and findings may advance theoretical research and inform practice.

5.1. Implications and contributions

In terms of theoretical implications, this research proposes three new variables, namely combination potential, combination intensity, and combination distance, to describe the combination characteristics of knowledge elements. It is found that, together with combination breadth, these variables have significant impacts on the combination frequencies between knowledge elements. Given that existing research on knowledge networks mainly focuses on binary networks and direct ties in networks (Wang et al., 2014; Zhang & Luo, 2021), with little consideration for edge weights or indirect ties, this research provides new insight into the knowledge combination theory and studies on knowledge networks. Specifically, the findings suggest that both direct and indirect ties as well as tie strengths among knowledge elements are important indicators of innovation because they could provide R&D clues and opportunities. Thus, our work enriches the variable systems of studies that take the perspective of knowledge combination to study innovation.

In terms of practical contributions, this research applies a machine learning method to patent data and proposes a novel method to identify technological opportunities. The proposed method can be applied in various contexts and domains, making it a versatile tool for technological opportunity discovery. It enriches the method library for utilizing patent data for technology forecasting. Unlike existing studies that primarily predict the occurrence of specific combinations of knowledge elements (Ren & Zhao, 2021; Yang et al., 2022; Zhang & Luo, 2020b), this research goes a step further by predicting the frequency of the combinations. This provides a more nuanced understanding of which combinations are likely to be more prevalent or impactful, enabling researchers and practitioners to prioritize their efforts more effectively.

5.2. Limitations and future research

This research has some limitations. First, the effectiveness of the proposed method is empirically tested using patent data in the field of AI and the field of IoT. The combination characteristics of knowledge elements may vary in different technical fields, so the applicability of the proposed method in other technical domains still needs to be examined. Second, the prediction accuracy of the proposed method relies on the training effectiveness of random forest models, which has certain requirements for the amount of data used for model training. Therefore, whether this method is suitable for scenarios with a small amount of data still needs to be verified. Future research can examine the robustness of the proposed method by using data from multiple domains and varying data volumes. Third, this research uses IPC codes to denote knowledge elements, which may fully capture the actual contents of specific technologies. Future research may replace them with technological keywords extracted from patent contents, which may more accurately reflect the relationships between technological elements and the knowledge map in a specific field. Additionally, the technological opportunity discovery is conducted based on patent data in this research. Overlapping patents and defensive patents may obscure true innovation trends and dilute the signal-to-noise ratio. Therefore, it is better to complement patent data with other data (e.g., scientific literature, market data, expert insights) for more robust opportunity identification.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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